

Amirjon Abdunayimov

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EDUCATION:

New York University — Tandon School of Engineering

Expected Graduation: May 2029

Bachelor of Science in Computer Science (Minor in Mathematics)

Relevant Coursework: Data Structures & Algorithms, Discrete Mathematics, Linear Algebra, Probability & Statistics

Academic Honors (Kingsborough CC): Jack Kent Cooke Transfer Scholarship Semifinalist, Honors Scholar, Phi Theta Kappa, Dean's List

WORK EXPERIENCE:

Boardwalk Labs — Software Engineering Intern (Part-Time)

June 2026 – Sep 2026

- Architected an event-driven **Go/Python microservices engine** for OctoPilot AI utilizing background worker queues and connection pooling, optimizing distributed task routing to cut end-to-end processing latency by **50%**.
- Engineered a fault-tolerant multi-agent execution pipeline using **Redis-backed cache-aside patterns**, TTL invalidation, and token-bucket rate limiting; integrated Prometheus tracking to handle long-running LLM stream failures.
- Designed an **ACID-compliant transactional ledger** in PostgreSQL for high-concurrency credit accounting, leveraging database indexing and strict isolation levels to eliminate race conditions and slash infrastructure API leaks by **28%**.

Sound of Earth — DevOps & Cloud Infrastructure Intern (Part-Time)

May 2026 – Aug 2026

- Architected and owned containerized CI/CD infrastructure using **Docker, GitHub Actions, and Linux**, automating multi-environment deployments to reduce production regressions by **40%** and slash **\$3.2K/year** in staging overhead.
- Engineered a **zero-downtime Supabase/PostgreSQL migration pipeline** for **138,000+ records**, restructuring legacy data across **8 normalized tables** and building **PostgREST-backed APIs** for web and mobile synchronization.
- Audited and hardened production infrastructure security across **JWT authentication, CORS, secrets management, and isolated Docker networking**, identifying and remediating **29 core exposure vectors**.

Kingsborough Community College Learning Center — Computer Science & Mathematics Tutor

March 2026 – Aug 2026

- Mentored **50+ students across 100+ sessions** in introductory programming and mathematics, translating complex concepts into clear, structured problem-solving strategies.
- Provided individualized tutoring in **C++, pre-calculus, algebra, and statistics**, adapting explanations and practice problems to students' skill levels and coursework.
- Guided students through **programming fundamentals, debugging, and code tracing**, helping them identify errors and develop more systematic approaches to solving problems.

PROJECTS:

sev0 — Autonomous AI Incident Investigation & Repair System | Python, Anthropic API, FastAPI, PostgreSQL, Docker, Prometheus, Loki

- Built an **autonomous AI software engineer for production incidents**, orchestrating **16 diagnostic tools** across Prometheus, Loki, Git, and Docker to trace failures to the root-cause service, file, symbol, and introducing commit.
- Designed a **4-service incident lab with 23 fault scenarios**, including **35%** adversarial and silent failures, to evaluate root-cause analysis across code, configuration, infrastructure, and business-metric regressions.
- Engineered a **sandboxed repair-and-verification pipeline** that reproduces failures before patching, runs regression checks on proposed fixes, enforces patch safety limits, and generates **draft GitHub PRs** for human review.

StateRelay — Cross-Platform Workspace State Synchronization | Go, TypeScript, Git, Linux, SQLite, mDNS, TLS

- Built a cross-platform synchronization system in **Go + TypeScript**, capturing **7+ workspace-state signals** across Git, editor, terminal, and browser context.
- Developed VS Code and browser integrations with a Go-based capture/restore pipeline, achieving complete workspace restoration across **Linux, macOS, and Windows**.
- Engineered **secure peer-to-peer transfer** using mDNS, mutual TLS, Ed25519 signatures, and SHA-256 integrity verification.

LEADERSHIP:

AI, Coding, and Systems (ACS) Club (Kingsborough CC) — Founder & President

March 2026 – June 2026

- Founded a computer science organization focused on **software engineering, AI, and systems programming**, leading technical workshops, project-based learning, and peer development sessions.

PROFESSIONAL SKILLS:

Programming Languages: Python, C++, Go, TypeScript, JavaScript, Dart, SQL

Frameworks & Data: FastAPI, React, Flutter, Node.js, PostgreSQL, SQLite

Tools & Infrastructure: Docker, GitHub Actions (CI/CD), Linux, Git, REST APIs, WebSockets, Prometheus